

Clinical Significance of the Hyperintense Reperfusion Marker and Ocular Gadolinium Leakage After Thrombectomy

A Retrospective ETIS Study

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Abstract

Background and Objectives

Despite successful recanalization after mechanical thrombectomy for acute ischemic stroke, a substantial proportion of patients experience early neurologic deterioration or infarct progression. Early blood-brain barrier disruption, visualized on post-contrast fluid-attenuated inversion recovery (pcFLAIR) imaging as hyperintense acute reperfusion marker (HARM) or gadolinium leakage in ocular structures (GLOS), has been proposed as a marker of reperfusion injury, but its determinants and prognostic significance in thrombectomy cohorts remain uncertain. We aimed to evaluate whether HARM and GLOS are associated with poor clinical outcomes after mechanical thrombectomy. Secondary objectives included their association with poor radiologic outcomes and the identification of baseline and procedural predictors.

Methods

We retrospectively analyzed data from the French Endovascular Treatment in Ischemic Stroke (ETIS) registry (2010–2021), including consecutive patients admitted to Versailles and Foch Hospital. All patients underwent mechanical thrombectomy at Foch Hospital and pcFLAIR imaging within 4–48 hours postgadolinium administration. Independent predictors of HARM and GLOS were identified through multivariable logistic regression, and their associations with early neurologic deterioration, poor 3-month functional outcome, infarct growth, and hemorrhagic transformation (HT) outcomes were evaluated.

Results

Among 229 patients (mean age 69.7 ± 14.9 years, 52.0% female), HARM was observed in 48.9% and GLOS in 41.9%. Overall, 60.3% of patients had either HARM or GLOS, including 29.3% with both markers, 19.7% with isolated HARM, and 11.4% with isolated GLOS. HARM was independently associated with early neurologic deterioration (adjusted odds ratio [aOR] 3.45; 95% CI 1.33–8.95) and infarct growth (aOR 2.46; 95% CI 1.22–4.95), but not with HT or 3-month functional outcome. GLOS was not independently associated with clinical or radiologic outcomes. Older age, higher creatinine, and a shorter interval between the first and second FLAIR were common predictors of both HARM and GLOS, whereas single-pass recanalization was an independent predictor of HARM only.

Discussion

In this large thrombectomy cohort, HARM was independently associated with early neurologic deterioration and infarct progression after thrombectomy, supporting its role as an imaging marker of early blood-brain barrier disruption. GLOS, although frequent, was not

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Glossary

AIC = Akaike information criterion; **aOR** = adjusted odds ratio; **ASITN/SIR** = American Society of Interventional and Therapeutic Neuroradiology/Society of Interventional Radiology; **DWI** = diffusion-weighted imaging; **END** = early neurological deterioration; **ETIS** = Endo-vascular Treatment in Ischemic Stroke registry; **EVT** = endovascular treatment; **FLAIR** = fluid-attenuated inversion recovery; **GLOS** = gadolinium leakage in ocular structures; **HARM** = hyperintense acute reperfusion marker; **HT** = hemorrhagic transformation; **mRS** = modified Rankin Scale; **MT** = mechanical thrombectomy; **mTICI** = modified Thrombolysis in Cerebral Infarction; **NIHSS** = NIH Stroke Scale; **pcFLAIR** = post-contrast FLAIR; **VIF** = variance inflation factor.

independently associated with clinical or radiologic outcomes. Prospective studies with early and serial imaging are required to clarify the prognostic relevance of HARM.

Trial Registration Information

NCT03776877 (ETIS registry).

Introduction

Despite the success of mechanical thrombectomy (MT) in achieving vessel recanalization in more than 70% of acute ischemic strokes, nearly half of treated patients continue to experience infarct growth, hemorrhagic transformation, or poor functional recovery.¹ Ineffective recanalization after mechanical thrombectomy may partly reflect early microvascular and barrier injury occurring within the hyperacute reperfusion phase. On post-contrast fluid-attenuated inversion recovery (pcFLAIR) imaging, hyperintense acute reperfusion marker (HARM)² and gadolinium leakage in ocular structures (GLOS)³ correspond to gadolinium extravasation into the subarachnoid sulci or ocular compartments detectable as early as 5 minutes after contrast administration and subsequently redistributing within the CSF. Thus, HARM and GLOS are considered imaging manifestations of early permeability failure and can be readily identified on pcFLAIR acquired in routine clinical practice, even when imaging is performed beyond the hyperacute phase.⁴

Two systematic reviews have highlighted substantial heterogeneity in reported prevalence due to methodology assessment for HARM and GLOS, complicating the interpretation of their clinical relevance.^{5,6} HARM has been variably associated with lesion volume expansion, early neurologic deterioration, and hemorrhagic transformation, but its prognostic links remain inconsistent across small, heterogeneous cohorts. Although small cohort studies have suggested a worse early neurologic prognosis in HARM-positive thrombectomy patients,⁷ robust prognostic factors for HARM await validation in larger cohorts. By contrast, GLOS likely represents physiologic excretion of gadolinium,⁸ with persistent enhancement suggesting localized blood-ocular barrier dysfunction related to cardiovascular risk factors⁹; however, no systematic evaluation of GLOS post-thrombectomy exists.

Recognizing these gaps, our primary objective was to assess the association of HARM and GLOS with poor clinical outcomes after mechanical thrombectomy. Secondary objectives

were to examine their association with poor radiologic outcomes and the identification of baseline and procedural predictors.

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

The study was conducted in accordance with institutional ethical standards and the Declaration of Helsinki. All patients included in the Endovascular Treatment in Ischemic Stroke registry (ETIS; NCT03776877), or their legal representatives, provided oral informed consent for inclusion in the registry and for the use of anonymized clinical and imaging data for research purposes. Given the retrospective nature of the present analysis using prospectively collected registry data, no additional study-specific consent was required according to national regulations. The study protocol was approved by the appropriate institutional review board/ethics committee (2017-A03457-46, CPP Sud-EST 1).

Study Population

We included all consecutive patients admitted between January 2010 and January 2021 at Versailles Hospital (Primary Stroke Center) and Foch Hospital (Comprehensive Stroke Center). All patients underwent mechanical thrombectomy at Foch Hospital. Eligible patients had baseline MRI with gadolinium injection and precontrast FLAIR before mechanical thrombectomy, followed by pcFLAIR imaging between 4 and 48 hours after contrast administration. Patients with subarachnoid hemorrhage or uninterpretable MRI scans were excluded.

Clinical and Neuroradiologic Data Collection

Demographics (age, sex, and risk factors), clinical data, and outcomes (baseline and 24-hour NIH Stroke Scale [NIHSS] score, 3-month modified Rankin Scale [mRS] score, and time metrics) were prospectively collected in the ETIS Registry by

certified neurologists. European Cooperative Acute Stroke Study scale for hemorrhagic transformation (HT), and procedural data such as first-line technique, modified Thrombolysis in Cerebral Infarction (mTICI) score, and number of passes, were prospectively collected in the ETIS Registry by an expert neuroradiologist. Recanalization status was assessed using the mTICI scale, with successful recanalization defined as mTICI $\geq 2b$. The American Society of Interventional and Therapeutic Neuroradiology/Society of Interventional Radiology (ASITN/SIR) collateral grading scale were retrospectively assessed by 2 expert neuroradiologists in a previous study.¹⁰ Poor collateral status was dichotomized for ASITN/SIR grades 0–2.¹¹

Two independent raters (C.B. and G.V.), blind to clinical data, assessed the baseline FLAIR images and pcFLAIR at follow-up for any hyperintensity in subarachnoid spaces or in ocular structures. Especially, HARM was defined as a new pcFLAIR enhancement in supratentorial subarachnoid spaces compared with baseline FLAIR. Focal (sulcal or lobar) or diffuse patterns were also recorded. GLOS was defined as pcFLAIR enhancement in ocular structures, including anterior and/or posterior chambers, and/or perineural space of the optic nerves. Discrepancies were resolved by a senior neuroradiologist (A.C.). Interobserver and intraobserver agreement (Cohen κ) was calculated for HARM and GLOS. FLAIR vascular hyperintensity was rated after FVH-ASPECT gradation¹² on precontrast FLAIR images as circular or serpentine hyperintensities in brain parenchyma or on the cortical surface bordering the subarachnoidal space and dichotomized as poor for scores below 4. Diffusion-weighted imaging (DWI) volume at baseline and MRI follow-up were manually segmented by an expert radiologist (J.G.). All MRI examinations were performed on either 1.5 T or 3 T scanners. Subarachnoid hemorrhage and hemorrhagic transformation were assessed on gradient-echo MRI or noncontrast CT.

Clinical outcomes included early neurologic deterioration (END) defined as an increase of 4 or more points on the 24-hour NIHSS score and a 3-month mRS score greater than 2. Neuroradiologic outcomes included absolute infarct growth, defined as an increase of more than 3 mL to account for segmentation variability,¹³ and presence of any HT.

Statistical Analysis

Continuous variables were summarized as means with standard deviations, while categorical variables were described as counts and percentages. Univariate logistic regression analyses were performed to identify potential risk factors associated with HARM and GLOS, as well as with outcomes. Variables with p -value < 0.20 in these analyses and less than 10% missing data were considered candidates for multivariable models. Multicollinearity among candidate variables was assessed using the variance inflation factor (VIF). Final multivariable logistic regression models for HARM, GLOS, and outcomes were constructed using a stepwise variable selection approach guided by the Akaike information criterion (AIC), optimizing

parsimony. Primary models were prespecified to include either HARM or GLOS (but not both). As sensitivity analyses for HARM and GLOS, we repeated the stepwise AIC selection without excluding either marker, allowing both HARM and GLOS to enter the model. Adjusted odds ratios (aORs) with corresponding 95% CIs were reported. All statistical analyses were performed using Python (version 3.13) with the Statsmodels library, and statistical significance was defined as $p < 0.05$.

Data Availability

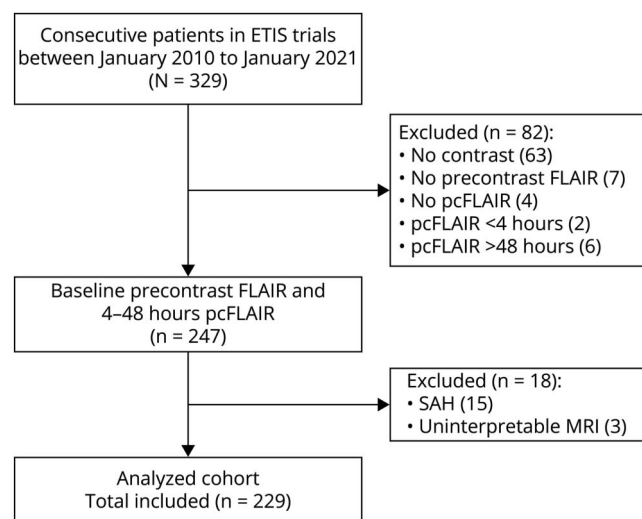
Data could be made available upon request to the principal investigators.

Results

Patient Inclusion

A total of 329 consecutive patients enrolled in the ETIS trials in the 2 centers between January 2010 and January 2021 were retrospectively screened for eligibility (Figure 1). Of these, 82 patients were excluded for the following reasons: absence of contrast administration ($n = 63$), no available precontrast FLAIR MRI at admission ($n = 7$), missing follow-up pcFLAIR ($n = 4$), pcFLAIR acquired < 4 hours from baseline ($n = 2$), or > 48 hours ($n = 6$). This yielded 247 patients with both baseline precontrast FLAIR and 4–48 hours follow-up pcFLAIR sequences. An additional 18 patients were excluded because of either a hemorrhagic appearance in the subarachnoid space detected on gradient-echo MRI or CT imaging ($n = 15$) or uninterpretable MRI data ($n = 3$). A total of 229 patients were included in the final analyzed cohort.

Figure 1 Flowchart of the Study



ETIS = Endovascular Treatment in Ischemic Stroke; FLAIR = fluid-attenuated inversion recovery; pcFLAIR = post-contrast FLAIR; SAH = subarachnoid hemorrhage.

Descriptive Characteristics of HARM and GLOS Presentations

In the overall cohort ($n = 229$), HARM was present in 112 patients (48.9%) and GLOS in 93 of 222 evaluable patients (41.9%). Overall, 138 (60.3%) patients had either HARM or GLOS, including 67 (29.3%) with both markers, 45 (19.7%) with isolated HARM, and 26 (11.4%) with isolated GLOS.

HARM presentation was categorized as focal or diffuse. Focal enhancement occurred in 64 of these patients (57.1%), with 11 (9.8%) exhibiting sulcal enhancement in 1 or 2 adjacent sulci. Diffuse enhancement was observed in 81 patients (72.3%), comprising 47 (42.0%) with intense enhancement and 53 (47.3%) with a fuzzy pattern. Bilateral HARM was present in 38 cases (34.0%). Concomitant ventricular enhancement occurred in 6 (5.4%) and posterior fossa enhancement in 56 (50.0%) of HARM-positive patients. Notably, in 10 cases (8.9%) where enhancement was confined to the posterior fossa, the finding was not classified as HARM because of lack of supratentorial involvement.

GLOS assessment was hindered by ocular motion artifacts in 87 of 229 patients (38.0%), of whom 7 cases (3.1%) were deemed uninterpretable and excluded. Enhancement was detected in the anterior chamber in 58 patients (62.4%), the posterior chamber in 68 (73.1%), and the optic nerve region in 84 (90.3%) of GLOS-positive cases. GLOS was bilateral in 82 patients (88.2%), and in 19 cases (20.4%) enhancement was limited to the subretinal space and/or the perineural optic nerve sheath.

Interobserver and intraobserver agreements were excellent for HARM ($\kappa = 0.88$ and 0.91 , respectively), substantial for GLOS ($\kappa = 0.72$ and 0.76). The main discrepancies arose in cases with a fuzzy HARM pattern or isolated to the subretinal space and/or the perineural optic nerve sheath, where these features were the sole components present. Representative cases are presented in eFigures 1–3.

Cohort Characteristics and Univariate Associations

Compared with patients without HARM (Table 1), those with HARM were significantly older (73.5 ± 14.4 vs 66.0 ± 14.6 years, $p < 0.001$), more frequently had hypertension (65.8% vs 50.9%, $p = 0.02$), higher admission NIHSS scores (16.1 ± 6.8 vs 13.7 ± 6.8 , $p < 0.01$), and elevated creatinine levels (85.8 ± 25.1 vs 76.4 ± 20.3 $\mu\text{mol/L}$, $p < 0.01$). Atrial fibrillation was also more frequent in the HARM group (25.8% vs 14.0%, $p = 0.047$). On baseline imaging, patients with HARM had shorter time intervals between baseline and follow-up MRI (20.9 ± 6.3 vs 24.7 ± 6.1 hours, $p < 0.001$) and presented larger DWI lesion volumes at admission (46.1 ± 50.8 vs 30.8 ± 42.7 mL, $p = 0.02$). Procedurally, HARM was associated with lower rates of first-pass thrombectomy (25.9% vs 45.3%, $p = 0.006$), although overall rates of successful recanalization were similar.

HARM was also significantly associated with GLOS (62.6% vs 22.6%, $p < 0.001$). Clinically, patients with HARM

experienced higher rates of END (19.1% vs 6.0%, $p < 0.01$). They were more likely to show absolute infarct growth greater than 3 mL (61.1% vs 40.5%, $p < 0.01$). Moreover, hemorrhagic transformation tended to be more frequent in the HARM group (66.1% vs 54.7%, $p = 0.08$). Poor 3-month functional outcome (mRS >2) was observed more often in patients with HARM (57.0% vs 31.8%, $p < 0.001$).

Compared with patients without GLOS (Table 2), those with GLOS were significantly older (74.2 ± 14.1 vs 66.9 ± 14.3 years, $p < 0.001$) and had higher creatinine levels (87.4 ± 26.1 vs 76.4 ± 19.1 $\mu\text{mol/L}$, $p < 0.001$). A trend toward higher prevalence of hypertension (65.2% vs 53.5%, $p = 0.08$), dyslipidemia (40.4% vs 29.4%, $p = 0.09$), and lower prevalence of active smoking (19.0% vs 29.7%, $p = 0.09$) was also observed in the GLOS group. Patients with GLOS showed longer delays from contrast injection to groin puncture (1.4 ± 0.6 vs 1.1 ± 0.6 hours, $p < 0.01$) and shorter delays from baseline to follow-up MRI (20.5 ± 5.9 vs 24.9 ± 5.9 hours, $p < 0.001$). GLOS was strongly associated with HARM (82.1% vs 42.0%, $p < 0.001$). No outcome was significantly associated with GLOS, despite a trend toward more frequent poor 3 months (52.9% vs 39.2%, $p = 0.05$).

Independent Predictors of HARM and GLOS

Multivariate analyses identified independent predictors of HARM and GLOS separately (Table 3). Older age and higher creatinine levels independently increased the odds of detecting both HARM (age: aOR 1.05 [95% CI, 1.02–1.07], $p < 0.01$; creatinine: aOR 1.02 [1.00–1.03], $p = 0.02$) and GLOS (age: aOR 1.04 [1.01–1.06], $p = 0.01$; creatinine: aOR 1.02 [1.00–1.03], $p = 0.04$). In addition, a shorter interval between baseline and follow-up FLAIR imaging was significantly associated with increased detection of both markers (per hour decrease: HARM, aOR 0.89 [0.84–0.94], $p = 0.001$; GLOS, aOR 0.85 [0.80–0.92], $p < 0.001$). Single-pass recanalization independently reduced the odds of HARM (aOR 0.47 [0.24–0.92], $p = 0.03$), while a longer delay from contrast injection to groin puncture showed a trend toward increased GLOS detection (aOR 1.67 [0.97–2.88], $p = 0.06$).

In sensitivity analyses, GLOS strongly predicted HARM occurrence (aOR 4.72 [2.17–10.26], $p < 0.001$), and reciprocally, HARM significantly predicted GLOS (aOR 3.67 [1.72–7.83], $p < 0.001$). In these models, the effects of creatinine and the first-to-second FLAIR interval for HARM, and of age for GLOS, attenuated and were no longer statistically significant (eTable 1), consistent with shared variance with the imaging markers rather than collinearity ($\text{VIF}_{\text{HARM}} = 1.3$, $\text{VIF}_{\text{GLOS}} = 1.4$).

Prognostic Value of HARM and GLOS for Clinical and Imaging Outcomes

Regarding clinical and radiologic outcomes (Table 4), the presence of HARM significantly predicted END (aOR 3.45; 95% CI 1.33–8.95; $p < 0.01$) and absolute infarct growth (aOR 2.46; 95% CI 1.22–4.95; $p < 0.01$). By contrast, HARM

Table 1 Baseline Characteristics and Univariable Associations With HARM

Independent variables	All (n = 229)	HARM– (n = 117)	HARM+ (n = 112)	Univariable regression	
				OR (95% CI)	p Value
Age	69.7 ± 14.9	66.0 ± 14.6	73.5 ± 14.4	1.04 (1.02–1.06)	<0.001 ^a
Female sex	119 (52.0)	62 (53.0)	57 (50.9)	0.92 (0.55–1.54)	0.75
History of hypertension ^b	132 (58.1)	59 (50.9)	73 (65.8)	1.86 (1.09–3.17)	0.02 ^a
Diabetes ^b	32 (14.3)	13 (11.6)	19 (17.1)	1.57 (0.74–3.36)	0.24
Dyslipidemia ^b	76 (34.4)	33 (29.5)	43 (39.4)	1.56 (0.89–2.73)	0.12
Active smoking ^c	55 (26.3)	31 (28.7)	24 (23.8)	0.77 (0.42–1.44)	0.42
Atrial fibrillation ^d	36 (19.8)	13 (14.0)	23 (25.8)	2.145 (1.01–4.56)	0.047 ^a
Antiaggregation therapy	68 (29.7)	30 (25.6)	38 (33.9)	1.49 (0.84–2.63)	0.17
Anticoagulation therapy	43 (18.8)	17 (14.5)	26 (23.2)	1.78 (0.91–3.50)	0.1
NIHSS at admission	14.9 ± 6.9	13.7 ± 6.8	16.1 ± 6.8	1.05 (1.01–1.10)	<0.01 ^a
Creatinine level (μmol/L) ^b	81.0 ± 23.2	76.4 ± 20.3	85.8 ± 25.1	1.02 (1.006–1.03)	<0.01 ^a
IV alteplase (rtPA) ^b	94 (41.2)	53 (45.3)	41 (36.9)	0.707 (0.42–1.20)	0.2
LVO	161 (70.3)	83 (70.9)	78 (69.6)	0.94 (0.53–1.66)	0.83
DWI volume at admission (mL) ^b	38.4 ± 47.4	30.8 ± 42.7	46.1 ± 50.8	1.007 (1.001–1.01)	0.02 ^a
Fazekas score					
0	26 (11.4)	16 (13.7)	10 (8.9)	1.0	0.06
1	108 (47.2)	57 (48.7)	51 (45.5)	0.88 (0.52–1.48)	
2	62 (27.1)	34 (29.1)	28 (25.0)	0.81 (0.45–1.46)	
3	33 (14.4)	10 (8.5)	23 (20.5)	2.77 (1.25–6.12)	
Poor FVH ^c	70 (32.3)	34 (30.6)	36 (34.0)	1.17 (0.66–2.06)	0.6
Poor collateral status ^d	131 (66.8)	67 (67.7)	64 (66.0)	0.93 (0.51–1.68)	0.8
Single pass ^c	76 (35)	45 (41.7)	31 (28.4)	0.56 (0.32–0.98)	0.04 ^a
No. of passes ^c	2.7 ± 2.1	2.6 ± 2.2	2.8 ± 1.9	1.07 (0.94–1.2)	0.3
Recanalization	192 (83.8)	100 (85.5)	92 (82.1)	0.78 (0.39–1.58)	0.49
GLOS ^a	93 (41.9)	26 (22.6)	67 (62.6)	5.73 (3.19–10.31)	<0.001 ^a
Delay from onset to contrast (h)	2.2 ± 2.2	2.1 ± 1.1	2.4 ± 2.9	1.07 (0.93–1.23)	0.36
Delay from contrast to puncture (h)	1.2 ± 0.6	1.2 ± 0.7	1.2 ± 0.5	0.94 (0.61–1.46)	0.79
Delay from puncture to recanalization (h) ^b	1.0 ± 0.6	0.9 ± 0.7	1.0 ± 0.6	1.23 (0.82–1.84)	0.3
Delay from onset to recanalization (h) ^b	4.4 ± 2.3	4.3 ± 1.6	4.5 ± 3.0	1.05 (0.93–1.19)	0.4
Delay between first to follow-up FLAIR (h)	22.8 ± 6.5	24.7 ± 6.1	20.9 ± 6.3	0.90 (0.85–0.94)	<0.001 ^a
Dose gadolinium (mL/kg) ^d	0.1 ± 0.03	0.1 ± 0.03	0.1 ± 0.04	0.50 (0.0–14311.3)	0.89
Follow-up MRI (3T) ^b	192 (86.5)	97 (85.8)	95 (87.2)	1.1 (0.5–2.4)	0.77
END ^b	28 (12.3)	7 (6.0)	21 (19.1)	3.71 (1.51–9.12)	<0.01
Absolute infarct growth ^b	111 (50.7)	45 (40.5)	66 (61.1)	2.31 (1.34–3.96)	<0.01
Any HT	138 (60.3)	64 (54.7)	74 (66.1)	1.61 (0.95–2.75)	0.08
Poor mRS at 3 mo ^c	96 (44.2)	35 (31.8)	61 (57.0)	2.84 (1.63–4.95)	<0.001 ^a

Abbreviations: DWI = diffusion-weighted imaging; END = early neurologic deterioration; FLAIR = fluid-attenuated inversion recovery; FVH = FLAIR vascular hyperintensity; GLOS = gadolinium leakage in ocular structures; HARM = hyperintense acute reperfusion marker; HT = hemorrhagic transformation; LVO = large vessel occlusion; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale; OR = odds ratio; rtPA = recombinant tissue plasminogen activator (IV thrombolysis).

Continuous variables are presented as mean ± SD; categorical variables are expressed as counts (percentages).

^ap < 0.05.

^bMissing data <5%.

^cMissing data <10%.

^dMissing data ≥10%.

was not retained in the final models for any hemorrhagic transformation and, although retained in the model for poor 3-month functional outcome, was not significantly associated with this outcome (mRS >2). GLOS was not a significant predictor for any of the evaluated clinical or radiologic outcomes in multivariate models (see eTables 2–5 for full models).

Discussion

In this 10-year thrombectomy cohort, HARM was independently associated with END and with absolute infarct growth, whereas GLOS showed no independent prognostic value across clinical or radiologic outcomes. Regarding determinants, older age, higher creatinine, and a shorter interval between the first and second FLAIR were common predictors of both HARM and GLOS, while single-pass recanalization was independently associated with a lower occurrence of HARM, directionally concordant with reports linking multiple passes to HARM at 24 hours.⁷

Our data, consistent with prior thrombectomy studies, underscore that HARM reliably captures early barrier injury, translating to worse early neurologic trajectories. Gupta et al.¹⁴ previously reported that the absence of HARM independently predicted ≥ 8 -point NIHSS improvement at 24 hours, mirroring our observation that HARM predicts early worsening rather than recovery. Similarly, Luby et al.⁷ found higher 24-hour NIHSS when HARM was present, further supporting the link between barrier disruption and early clinical course. Extending to tissue outcomes, our association between HARM and lesion expansion aligns with literature tying barrier disruption to larger follow-up infarcts: Renú et al.¹⁵ reported that vessel wall enhancement and severe CSF-barrier disruption tracked with increased final infarct volume after thrombectomy; however, their definition combined sulcal gadolinium leakage with hemorrhage, limiting direct comparisons with isolated HARM sign. Conversely, Luby et al.¹⁶ found no univariate association between HARM and infarct growth in a relatively small endovascular treatment (EVT) cohort pointing to possible power limitations and/or methodological differences (scan timing, operational definitions). Turning to hemorrhagic transformation, the association with HARM remains inconsistent in stroke studies. In thrombectomy cohorts, the only study to adjust for key covariates⁷ did not find an independent association; consistent with our model, in which HARM was not retained for any HT. In adjusted analyses, HT risk in our cohort was primarily explained by infarct burden and treatment/procedural factors, rather than by HARM itself. Although intravenous thrombolysis has been implicated in blood-brain barrier disruption through matrix metalloproteinase activation,^{17,18} its association with HARM remains inconsistent across studies.⁶ Only a minority of studies reported a significant association, and none of the EVT-specific cohorts demonstrated such a relationship. In EVT-treated populations, HARM may primarily

reflect mechanisms related to endovascular therapy, with the effects associated with intravenous thrombolysis potentially outweighed, and without clear evidence of a synergistic effect between the 2 treatments.

Historically, Warach and Latour posited that CSF enhancement on postcontrast FLAIR marks deleterious reperfusion from microvascular injury,² yet this association has not been reproduced in unselected stroke cohorts. The EVT studies of Luby et al. suggest that HARM is not consistently associated with perfusion profiles in EVT-treated patients. While no relationship was observed with residual hypoperfusion¹⁹ or postischemic hyperemia,¹⁶ the latest large EVT cohort²⁰ with serial imaging showed a time-dependent association with incomplete reperfusion, present early after EVT, absent at 24 hours, and re-emerging at later time points, highlighting the temporal complexity of these imaging patterns. In our cohort, collateral status, classically viewed as a key determinant of microvascular perfusion, was likewise not associated with HARM. In the rapid, controlled reperfusion setting of thrombectomy, procedural load may therefore outweigh reperfusion or collateral status in driving HARM incidence.

Recent imaging work offers a mechanistic explanation for our findings. Quantitative metrics of perivascular fluid transport show reduced glymphatic activity ipsilateral to the infarct in the acute phase, consistent with impaired CSF-interstitial exchange after ischemia reperfusion.^{21,22} Within this framework, poststroke glymphatic dysfunction can initiate a deleterious cycle: Impaired CSF-interstitial exchange permits local accumulation of neurotoxic solutes, which damage the vascular endothelium, thereby exacerbating blood-brain barrier leakage and further perpetuating clearance failure.²³ The multipass thrombectomy context may exacerbate this vicious cycle by intensifying barrier disruption, diminishing glymphatic clearance, and ultimately potentiating infarct expansion. The persistence and spread of HARM may signal glymphatic dysfunction.

Although gadolinium in CSF may leak from the blood-brain barrier, human tracer studies,^{8,24} using heavily T2-weighted FLAIR, also indicate that physiologic excretion of gadolinium from ocular structures can drain into the basal cisterns and subsequently spread through the subarachnoid spaces, even in the absence of brain pathology or renal impairment. Consequently, symmetric and diffuse CSF hyperintensity can blur the boundary between HARM and GLOS, underscoring the need for time-resolved studies to define gadolinium washout kinetics and the optimal pcFLAIR window to distinguish these entities.^{4,8,24}

Interestingly, Förster et al.²⁵ reported an association between asymmetric GLOS and internal carotid artery stenosis or occlusion and a parallel may exist with the link between multiple thrombectomy passes and HARM, consistent with localized ischemia-reperfusion stress. Beyond their distinct

Table 2 Baseline Characteristics and Univariable Associations With GLOS

Independent variables	All (n = 222)	GLOS– (n = 129)	GLOS+ (n = 93)	Univariable regression	
				OR (95% CI)	p Value
Age	70.0 ± 14.6	66.9 ± 14.3	74.2 ± 14.1	1.04 (1.02–1.06)	<0.001 ^a
Female sex	115 (51.8)	65 (50.4)	50 (53.8)	1.145 (0.67–1.95)	0.62
History of hypertension ^b	129 (58.4)	69 (53.5)	60 (65.2)	1.63 (0.94–2.83)	0.08
Diabetes ^b	31 (14.3)	16 (12.5)	15 (16.9)	1.42 (0.66–3.04)	0.37
Dyslipidemia ^b	74 (34)	37 (29.4)	36 (40.4)	1.63 (0.92–2.89)	0.09
Active smoking ^c	51 (25.2)	35 (29.7)	16 (19.0)	0.56 (0.29–1.09)	0.09
Atrial fibrillation ^d	36 (20.1)	17 (16.7)	19 (24.7)	1.64 (0.79–3.4)	0.19
Antiaggregation therapy	65 (29.3)	35 (27.1)	30 (32.3)	1.28 (0.71–2.29)	0.4
Anticoagulation therapy	43 (19.4)	23 (17.8)	20 (21.5)	1.26 (0.65–2.47)	0.49
NIHSS at admission	14.8 ± 6.9	14.4 ± 7.1	15.3 ± 6.6	1.02 (0.98–1.06)	0.31
Creatinine level (μmol/L) ^b	81.0 ± 22.9	76.4 ± 19.1	87.4 ± 26.1	1.02 (1.01–1.04)	<0.001 ^a
IV alteplase (rtPA) ^b	89 (40.3)	55 (42.6)	34 (37.0)	0.79 (0.46–1.37)	0.4
LVO	156 (70.3)	90 (69.8)	66 (71.0)	1.06 (0.59–1.9)	0.85
DWI volume at admission (mL) ^b	37.1 ± 46.8	32.0 ± 42.6	44.0 ± 51.5	1.0 (0.99–1.01)	0.07
Fazekas score					
0	26 (11.7)	19 (14.7)	7 (7.5)	1.0	0.19
1	103 (46.4)	61 (47.3)	42 (45.2)	0.92 (0.54–1.57)	
2	60 (27.0)	34 (26.4)	26 (28.0)	1.08 (0.60–1.97)	
3	33 (14.9)	15 (11.6)	18 (19.4)	1.82 (0.87–3.84)	
Poor FVH ^c	68 (32.4)	39 (32.2)	29 (32.6)	1.02 (0.57–1.82)	0.96
Poor collateral status ^d	126 (66.7)	75 (65.8)	51 (68.0)	1.11 (0.59–2.06)	0.75
Single pass ^c	74 (35.2)	41 (33.3)	33 (37.9)	1.22 (0.69–2.18)	0.49
No. of passes ^c	2.7 ± 2.1	2.9 ± 2.3	2.4 ± 1.7	0.90 (0.78–1.04)	0.16
Recanalization	186 (83.8)	109 (84.5)	77 (82.8)	0.88 (0.43–1.81)	0.73
HARM	107 (48.2)	40 (31.0)	67 (72.0)	5.73 (3.19–10.31)	<0.001 ^a
Delay from onset to contrast (h)	2.3 ± 2.2	2.3 ± 2.5	2.2 ± 1.7	0.98 (0.86–1.11)	0.74
Delay from contrast to puncture (h)	1.2 ± 0.6	1.1 ± 0.6	1.4 ± 0.6	2.11 (1.27–3.49)	<0.01 ^a
Delay from puncture to recanalization (h) ^b	1.0 ± 0.7	1.0 ± 0.7	0.9 ± 0.6	0.79 (0.51–1.22)	0.3
Delay from onset to recanalization (h) ^b	4.4 ± 2.4	4.4 ± 2.6	4.4 ± 1.9	1.00 (0.89–1.12)	0.96
Delay between first to follow-up FLAIR (h)	23.0 ± 6.3	24.9 ± 5.9	20.5 ± 5.9	0.87 (0.82–0.92)	<0.001 ^a
Dose gadolinium (mL/kg) ^d	0.1 ± 0.03	0.1 ± 0.03	0.1 ± 0.03	0.0 (0.0–986.7)	0.5
Follow-up MRI (3T) ^b	186 (86.5)	107 (86.3)	79 (86.8)	1.05 (0.5–2.3)	0.9
END ^b	28 (12.7)	13 (10.2)	15 (16.3)	1.72 (0.78–3.82)	0.18
Absolute infarct growth ^b	111 (52.4)	66 (53.7)	45 (50.6)	0.88 (0.51–1.53)	0.66
Any HT	134 (60.4)	73 (56.6)	61 (65.6)	1.46 (0.84–2.54)	0.18
Poor mRS at 3 mo ^c	94 (44.8)	49 (39.2)	45 (52.9)	1.745 (1.0–3.05)	0.05

Abbreviations: DWI = diffusion-weighted imaging; END = early neurologic deterioration; FLAIR = fluid-attenuated inversion recovery; FVH = FLAIR vascular hyperintensity; GLOS = gadolinium leakage in ocular structures; HARM = hyperintense acute reperfusion marker; HT = hemorrhagic transformation; LVO = large vessel occlusion; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale; OR = odds ratio; rtPA = recombinant tissue plasminogen activator (IV thrombolysis).

Continuous variables are presented as mean ± SD; categorical variables are expressed as counts (percentages).

^ap < 0.05.

^bMissing data <5%.

^cMissing data <10%.

^dMissing data ≥10%.

associations, the convergent determinants we observed, older age, higher creatinine, and a shorter interval between the first and second FLAIR, suggest a vulnerability shared by HARM and GLOS, plausibly mediated by impaired CSF-interstitial exchange under ischemia-reperfusion. This common substrate may coexist with process-specific effects (e.g., multipass thrombectomy influencing HARM, but not GLOS). Although the close association with HARM may reflect partial overlap rather than 2 fully distinct entities, current evidence does not support GLOS as a prognostic marker in stroke. Noteworthy, higher-intensity signal in anterior chamber and the perineural space of the optic nerves has been linked to higher Fazekas scores in cohorts with or without brain metastases⁸ and in patients with stroke,^{3,9} which was not found in our study. Data on GLOS in thrombectomy cohorts remain limited, and further studies are needed.

HARM was analyzed as a binary variable. Other studies have attempted to grade HARM by intensity and extent, but both dimensions are highly time-dependent (intervals from contrast injection, reperfusion, and follow-up MRI), which is difficult to standardize retrospectively and remains challenging to define reproducibly. After initial sulcal leakage, contrast can redistribute within the CSF, often becoming more diffuse, supporting the notion that variable “washout” may be compatible with emerging data on glymphatic impairment after stroke.^{26,27} In practice, multiple patterns coexist, ranging from very intense signal in 1 or a few adjacent sulci, to contralateral or hemispheric involvement, to bilateral diffusion with

asymmetric (intense vs “fuzzy”) or symmetric appearance.⁶ Importantly, pcFLAIR signal intensity is a nonlinear function of gadolinium concentration, as shown by a phantom study,²⁸ which demonstrates that FLAIR is most sensitive to low concentrations, and that apparent “fuzzy” signal can arise from either highly diluted or heterogeneously concentrated gadolinium, complicating categorical schemes based on appearance alone.

Several limitations should be acknowledged, primarily related to the retrospective design relying on single time point pcFLAIR acquisitions performed as part of routine clinical practice and obtained between 4 and 48 hours after contrast administration and most often around 24 hours. Such a design precludes formal demonstration of temporal sequence and therefore does not support causal inference between HARM detection and outcomes. Although prior work suggests that sulcal gadolinium leakage occurs within the first 6 hours after stroke onset, our data demonstrate a strong independent association without formally establishing a predictive relationship. In addition, the variability in scan timing may introduce misclassification bias. Given the time-dependent nature of these phenomena, earlier imaging is more likely to capture pre-washout sulcal or ocular leakage, whereas later scans may miss transient leakage or detect only diffuse redistribution of uncertain origin, potentially reflecting either local blood-brain barrier disruption or physiologic ocular excretion. Moreover, HARM and HT evolve at different rates, and gadolinium washout kinetics vary considerably between patients. Single

Table 3 Independent Predictors of HARM and GLOS

Independent variables	aOR HARM (95% CI)	p Value	aOR GLOS (95% CI)	p Value
Age	1.05 (1.02–1.07)	<0.01 ^a	1.04 (1.01–1.06)	0.01 ^a
Dyslipidemia ^b	1.75 (0.88–3.45)	0.1	1.60 (0.78–3.25)	0.2
Active smoking ^c	—	—	0.85 (0.37–1.96)	0.7
Creatinine level (μmol/L) ^b	1.02 (1.00–1.03)	0.02 ^a	1.02 (1.00–1.04)	0.01 ^a
Delay from contrast to puncture (h)	—	—	1.67 (0.97–2.88)	0.06
Delay from first to follow-up FLAIR (h)	0.89 (0.84–0.94)	0.001 ^a	0.85 (0.80–0.92)	<0.001 ^a
DWI volume at admission (mL) ^b	1.01 (1.00–1.01)	0.07	1.01 (1.00–1.01)	0.08
Single pass ^c	0.47 (0.24–0.92)	0.03 ^a	—	—
Sensitivity analyses				
HARM	—	—	3.67 (1.72–7.83)	<0.001 ^a
GLOS ^b	4.72 (2.17–10.26)	<0.001 ^a	—	—

Abbreviations: AUCs = area under curve; aOR = adjusted odds ratio; DWI = diffusion-weighted imaging; FLAIR = fluid-attenuated inversion recovery; GLOS = gadolinium leakage in ocular structures; HARM = hyperintense acute reperfusion marker; NIHSS = NIH Stroke Scale.

Multivariable logistic regression with stepwise selection was used to identify baseline variables independently associated with HARM and GLOS. Variables were selected from univariable analyses ($p < 0.20$), excluding those with >10% missing data. Final models included 6 predictors for HARM ($n = 202$, 88.2%) and 7 for GLOS ($n = 189$, 82.5%), with AUCs of 0.79 and 0.83, respectively. Four variables were not retained after stepwise selection in the HARM model (history of hypertension, antiplatelet therapy, anticoagulation therapy, and admission NIHSS) and 1 variable in the GLOS model (history of hypertension). Global likelihood ratio tests were significant for both models ($p < 0.001$). Rows under “Sensitivity analyses” report the adjusted association when HARM or GLOS were not excluded from selection. Full sensitivity analyses models are available in the supplementary material. aORs and 95% CIs are reported for retained variables.

^a $p < 0.05$.

^bMissing data <5%.

^cMissing data <10%.

Table 4 Prognostic Impact of HARM and GLOS on Stroke Outcomes

Models	aOR HARM (95% CI)	p Value	aOR GLOS (95% CI)	p Value
END	3.45 (1.33–8.95)	0.01 ^a	1.86 (0.78–4.44)	0.16
Absolute infarct growth	2.46 (1.22–4.95)	0.01 ^a	—	—
Any HT	—	—	0.96 (0.48–1.95)	0.9
Poor mRS at 3 mo	2.40 (0.79–7.27)	0.12	1.01 (0.34–2.97)	0.98

Abbreviations: aOR = adjusted odds ratio; END = early neurologic deterioration; GLOS = gadolinium leakage in ocular structures; HARM = hyperintense acute reperfusion marker; HT = hemorrhagic transformation; mRS = modified Rankin Scale.

Each multivariable model was constructed using stepwise selection following univariable preselection ($p < 0.20$), excluding variables with $>10\%$ missing data. Outcomes models were prespecified to include either HARM or GLOS (but not both). aORs and 95% CIs are reported for HARM or GLOS when retained in the final model. Global likelihood ratio tests were significant for all models ($p < 0.001$). Full models are provided in the supplementary material.

^a $p < 0.05$.

time point imaging therefore, carries the risk of detecting HT after HARM has already resolved or identifying HARM before HT has emerged, potentially attenuating true associations. Nevertheless, the strength of these associations supports the need for prospective studies incorporating early and serial pcFLAIR imaging to clarify the temporal sequence and prognostic relevance of HARM. Although contrast-enhanced MRI is part of routine practice in the participating centers, a proportion of patients did not undergo gadolinium administration or lacked follow-up pcFLAIR imaging, leading to their exclusion. This may introduce selection bias and limit the generalizability of our findings. Although inter-rater agreement was high, image quality constraints were non-trivial; ocular motion artifacts were frequent, raising the possibility of residual misclassification. Finally, MRI protocols were not fully standardized over the 10-year inclusion period; differences in scanner manufacturers and acquisition parameters may have influenced pcFLAIR sensitivity and the visual appearance of HARM and GLOS, although some reports suggest this effect may be limited.^{29,30}

In conclusion, this retrospective cohort study demonstrates that HARM is independently associated with END and infarct growth after mechanical thrombectomy, bridging radiologic and clinical worsening. Conversely, GLOS was not independently associated with clinical or radiologic outcomes in our study. While this may suggest a predominantly physiologic mechanism, these findings should be interpreted with caution given the methodological limitations and the limited data available in thrombectomy cohorts. Further prospective studies with standardized imaging protocols, serial imaging, and quantitative analyses are warranted to better delineate the clinical utility and underlying mechanisms of HARM and GLOS.

Author Contributions

C. Bousfiha: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. G. Vinacci: major role in the acquisition of data. J. Gratioux: major role in the acquisition of data. I. Hmida: major role in the acquisition of data. M.-L. Chadenat:

major role in the acquisition of data. B. Lapergue: major role in the acquisition of data. A. Wang: major role in the acquisition of data. A. Consoli: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. F. Pico: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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